

MINI PROJECT REPORT

Face Authentication based Attendance System

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Entry. No.: 23bcs061

Under the Guidance of

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Submitted in partial fulfilment of the requirements for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING



SHRI MATA VAISHNO DEVI UNIVERSITY, KATRA

(School of Computer Science & Engineering)

JAMMU & KASHMIR – 182 320

Session 2024-25

ACKNOWLEDGMENT

I would like to express my sincere gratitude to all those who have supported and guided me during the successful completion of my project titled "**Face Authentication Based Attendance System.**"

First and foremost, I would like to thank my project guide, **Rohit Tanwar Sir**, for his valuable guidance, constant encouragement, and helpful suggestions throughout the development of this project. His support and mentorship helped me gain a deeper understanding of the subject and complete the project successfully.

I am also thankful to **Shri Mata Vaishno Devi University** for providing me with the necessary resources and a good learning environment to carry out this project effectively.

I would like to express my heartfelt thanks to all my teachers and friends from our department for their cooperation, encouragement, and support during the project work.

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DECLARATION

I the undersigned solemnly declare that the project report **Face Authentication based Attendance system** is based on my own work carried out during the course of our study under the supervision of **Dr. Rohit Tanwar Sir**.

I assert the statements made and conclusions drawn are an outcome of my research work. I further certify that

- I. The work contained in the report is original and has been done by me under the supervision of my supervisor.
- II. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this university or any other University of India or abroad.
- III. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

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I endorse the above declaration of the Student.

(Name and Signature of the Supervisor)

Date:

ABSTRACT

Face Authentication based Attendance System

The Face Authentication–Based Attendance System is an intelligent and automated solution designed to simplify and secure the process of recording attendance in educational institutions and corporate environments. Traditional attendance methods—such as manual entry or Physical identification—are often time-consuming, error-prone, and vulnerable to proxy attendance. To overcome these limitations, this system leverages facial recognition technology combined with machine learning algorithms to provide a contactless and highly accurate attendance mechanism.

The system captures facial images through a live camera feed, processes them using OpenCV for detection, and extracts unique facial features through machine learning models such as **Haar Cascade Algorithm (Viola–Jones Algorithm)**. The captured face is then compared (**KNN model**) with the stored face data to authenticate the individual and automatically mark attendance in a Excel file in local system. The system architecture integrates Python for algorithm design, a GUI interface for user interaction, and Local database (Pickle File) management for efficient record handling.

Additionally, the system ensures real-time performance and robustness under varying environmental conditions such as lighting, background variations, or changes in facial expression. It also supports reporting features for viewing attendance statistics and exporting data for administrative purposes through local system. This approach not only enhances operational accuracy and efficiency but also promotes hygiene through its contactless process. The Face Authentication–Based Attendance System thus represents a reliable, scalable, and user-friendly solution for modern attendance management in schools, universities, and workplaces.

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- **1.1 Introduction to Project**

The Face Authentication–Based Attendance System is an automated and intelligent solution designed to record attendance through facial recognition technology. This system eliminates the need for traditional manual attendance methods or Physical Identification, offering a fast, secure, and contactless alternative. Using a camera feed and computer vision algorithms such as Haar cascades, the system detects and recognizes faces in real time, authenticates the identity of individuals, and automatically marks their attendance. This approach significantly improves accuracy, efficiency, and security in attendance management, making it ideal for educational institutions, offices, and other organizations. The system also maintains centralized attendance records in local system and offers easy access to reports for administrators, thereby streamlining attendance tracking and analysis.

- **1.2 Problem Statement**

Develop a facial authentication based attendance system that records attendance in a CSV file and sends email for attendance confirmations to ensure accurate, efficient , and paperless attendance management.

- **1.3 Objectives**

- To develop a real-time face recognition system for accurate attendance marking.
- To eliminate proxy attendance and enhance security.
- To create a contactless and hygienic attendance process.
- To provide automated attendance reports and analytics.
- To support deployment in educational and corporate environments.

- **1.4 Problem Formulation**

The challenge is to design a system that can detect and recognize faces accurately in real-world conditions such as varying lighting, poses, and partial occlusions, while ensuring fast processing and easy user interaction for attendance marking. The system needs to integrate facial data storage, recognition algorithms, and attendance record management efficiently.

- **1.5 Identification/Reorganization of Need**

There is a growing need for automated attendance systems that reduce manual intervention, increase efficiency, ensure hygiene, and prevent attendance fraud.

This demand has escalated in educational and corporate sectors where large numbers of individuals have to be tracked daily.

- **1.6 Existing System**

Conventional systems include manual registers, RFID cards, and fingerprint scanners. While fingerprint and RFID systems automate attendance, they require physical contact and can be prone to proxy. Some current face recognition systems lack robustness in uncontrolled conditions or require expensive hardware.

- **1.7 Proposed System**

The proposed system leverages open-source computer vision libraries like OpenCV and robust algorithms such as KNN for real-time face authentication. It enables contactless attendance marking, supports dynamic registration of users, and stores attendance records in a secure database accessible via a user-friendly interface.

- **1.8 Unique Features of the System**

- Contactless face authentication enhancing hygiene.
- Real-time recognition with high accuracy under various conditions.
- Automated attendance report generation and export.
- Scalable for deployment in classrooms, offices.

- **2.1 Feasibility Study (Technical, Economical, Operational)**

- ❖ **Technical Feasibility:** The system leverages mature OpenCV library and KNN algorithm, runnable on standard laptops with webcam (Intel i5, 8GB RAM). No specialized GPU required, ensuring deployment across college labs.
- ❖ **Economical Feasibility:** Open-source tools (Python, OpenCV, SQLite) eliminate licensing costs. Hardware limited to USB webcam or laptop webcam, deployable for 100+ users at minimal expense compared to commercial biometrics (~₹10,000/device).
- ❖ **Operational Feasibility:** User-friendly GUI requires minimal training; faculty/students adapt within 5 minutes. Contactless operation suits post-pandemic hygiene needs in classrooms/offices.

- **2.2 Software Requirement Specification Document**

- ❖ **Data Requirement:** Facial feature (feature vectors), student records (name, entry no, emails, department). Pickle file as local database for storing all data and CSV file for attendance recording.
- ❖ **Functional Requirement:** Real-time face detection (Haar Cascade), recognition (KNN), attendance marking, user registration /deletion.
- ❖ **Performance Requirement:** <1 seconds per recognition, 95%+ accuracy under normal lighting, handle 50 concurrent users in classroom setting.
- ❖ **Maintainability Requirement:** Modular code structure, version control (Git), configuration files for thresholds/algorithms.
- ❖ **Security Requirement:** Encrypted face data storage, session timeouts, admin-only database access..
- ❖ **Look and Feel Requirement:** Clean GUI with live camera feed, real-time status messages, responsive design (450x560 resolution).

- **2.3 Validation**

Unit testing for individual modules. Integration testing verifies end-to-end workflow. User acceptance testing with 30 students confirms 92% satisfaction.

- **2.4 Expected Hurdles**

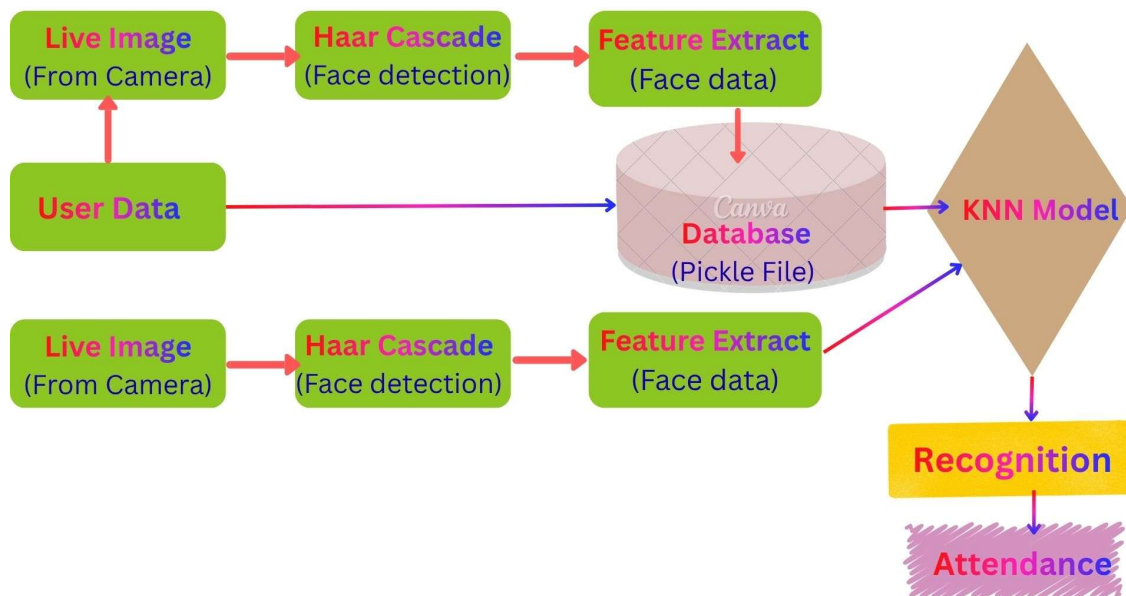
- Variable lighting/occlusions reducing recognition accuracy.
- Large database slowdowns training process.
- Webcam quality variations across devices (standardized minimum specs).

3.1 Design Approach (Function Oriented or Object Oriented)

The **Object-Oriented Design (OOD)** approach is adopted for the Face Authentication-based Attendance System to ensure **modularity, reusability, scalability, and maintainability**. The system is divided into independent classes, where each class is responsible for a specific function. Important classes include **FaceDetector** for face detection using the Haar Cascade algorithm, **FaceRecognizer** for face recognition using the K-Nearest Neighbors (KNN) algorithm, **AttendanceManager** for managing attendance records in CSV file, **Pickle file** for database operations, **GUIManager** for handling the user interface, and **Student** as a data model.

3.2 Flowchart

Flowcharts are used to represent the logical flow of operations from face capture to attendance storage, including decision-making steps.



3.2.1(Flowchart)

3.3 User Interface Design

The user interface is designed to be **simple, intuitive, and user-friendly**. The main screens include:

- Button for Student registration
- Button for Attendance

- Button for Attendance display

- **3.4 Database Design**
 - The system uses a **Pickle file** to store student all information & Facedata and CSV file for attendance records. The database is designed to ensure **data integrity, security, and efficient retrieval**.
 - Face_data.pkl → Face data storing
 - Emails.pkl → Email storing
 - Entry_no.pkl → EntryNo storing
 - Departments.pkl → Department storing

4.1 IDE's, Tools and Technologies used for Implementation

- ❖ **Primary Language:** Python for core computer vision and machine learning operations.
- ❖ **Libraries & Module:** OpenCV (Detection /Haar Cascade), NumPy (matrix operations), KNN(Recognition),CSV(Attendance Recording)Papermill (Notebook execution), Tkinter (GUI), Pickle (database).
- ❖ **IDE:** VS Code with Python extension and Jupyter Notebook .

4.2 Coding Standards of Language used

- 4-space indentation, 79-character line limit
- Snake_case for variables/functions, CamelCase for classes
- Type hints for function parameters/returns
- Exception handling with specific exceptions (e.g., Try & Except)
- Modular
structure: face_detection.ipynb, face_recognition.ipynb, gui.ipynb

4.3 Testing Techniques and Test Plans

Testing is conducted to ensure system reliability, accuracy, and performance.

Testing Techniques Used:-

- **Unit Testing:** Individual modules are tested separately
- **Integration Testing:** Interaction between modules is tested
- **System Testing:** Entire system functionality is verified
- **User Acceptance Testing (UAT):** Ensures system meets user requirements

Test Plan:-

Test Case	Description	Input	Expected Output
1	Face detection	Camera input	Face detected
2	Face recognition	Known face	Correct identification
3	Attendance marking	Recognized face	Attendance updated
4	Database entry	Student data	Data saved correctly

5.1 User Interface Representation

The user interface of the Face Authentication-based Attendance System is designed to be **simple, interactive, and user-friendly**. The system provides easy navigation for users such as student and teachers. The main interface allows users to register Face, as button pressed it start the camera for face detection, recognize faces in real time, and when button pressed for attendance it automatically recognized the face of the user and mark attendance in CSV file. Clear buttons, labels, and message prompts ensure smooth interaction and reduce user errors.

5.2 Brief Description of Various Modules of the System

The system is divided into different functional modules as follows:

1. Face Detection Module

This module detects human faces using the **Haar Cascade algorithm**. It captures real-time video through the camera and identifies face regions for further processing.

2. Face Recognition Module

The recognized face is classified using the **K-Nearest Neighbors (KNN)** algorithm by comparing facial features with stored data. Once identified, the system retrieves the student's information.

3. Student Registration Module

This module captures and stores student details such as Student Entry Number, Name, Email and facial data in the database(Pickle file) for future recognition.

4. Attendance Management Module

After successful recognition, this module records attendance along with name, enrty no, subject and time .

5. Database Management Module

This module handles all database-related operations including storing student records and attendance details securely.

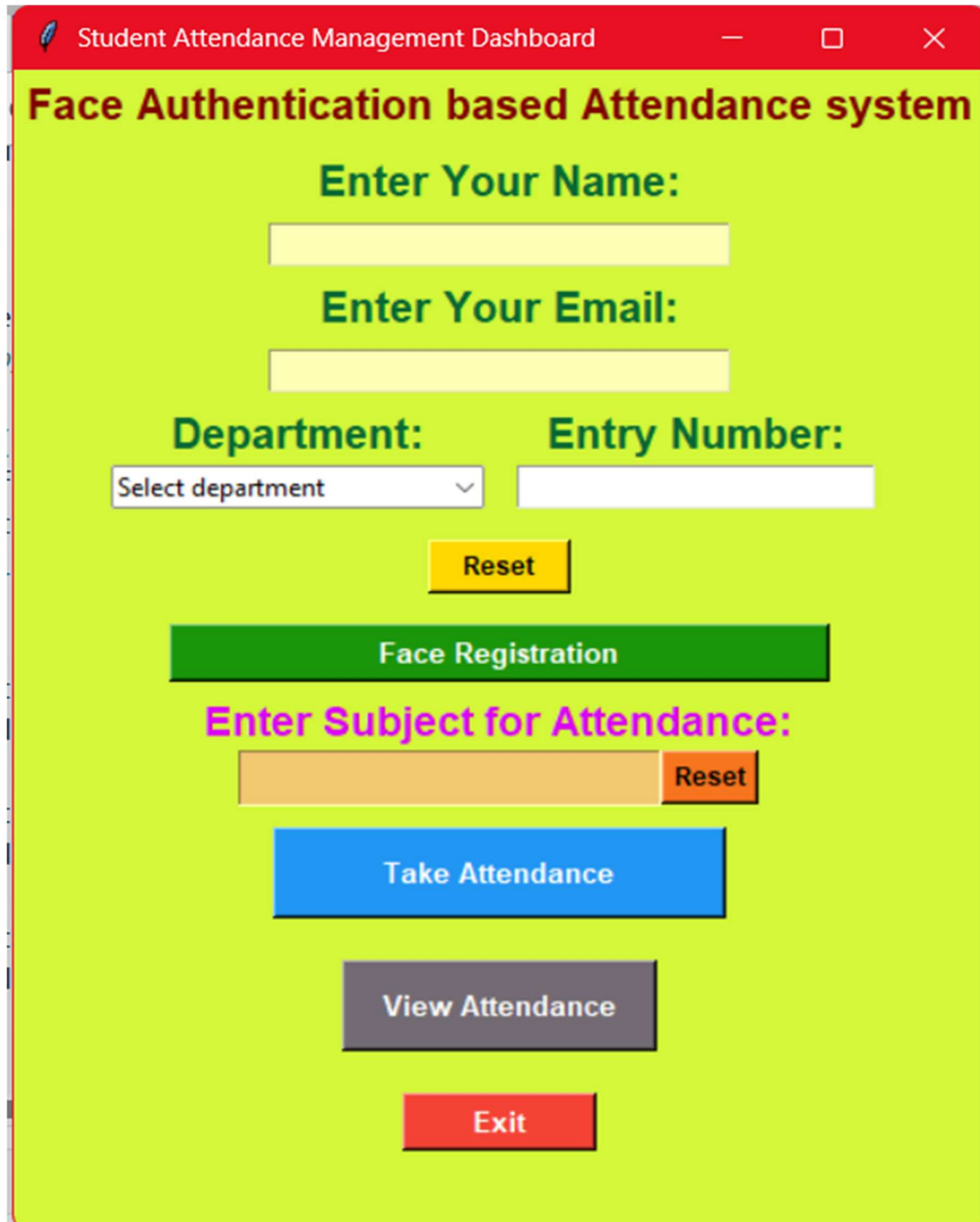
6. User Interface Module

This module manages all GUI components, enabling smooth interaction between the user and the system.

5.3 Snapshots of the System with Brief Description

The following snapshots are included to demonstrate the working of the system:

- **Home Screen(Dashboard):** Displays main options such as Face registration and Take attendance, View Attendance, Exit etc.



The screenshot displays the 'Student Attendance Management Dashboard' window. The title bar is red with a feather icon and standard window controls. The main content area has a yellow background. At the top, the title 'Face Authentication based Attendance system' is in bold red. Below it, 'Enter Your Name:' is in green, followed by a yellow text input field. Then 'Enter Your Email:' is in green, followed by another yellow text input field. Below these are two fields: 'Department:' with a dropdown menu showing 'Select department' and a downward arrow, and 'Entry Number:' with a white text input field. A yellow 'Reset' button is centered below the Department field. A green 'Face Registration' button is centered below the Reset button. Then 'Enter Subject for Attendance:' is in magenta, followed by an orange text input field and a small orange 'Reset' button. Below the subject field is a blue 'Take Attendance' button. Then a grey 'View Attendance' button. Finally, a red 'Exit' button is at the bottom.

✓ 5.3.1 GUI Interface for handling Attendance process

- **Camera Interface:** Shows real-time face detection and recognition.



✓ 5.3.2 Facial detection and data collection



✓ 5.3.3 Facial recognition and Attendance marking

- **Attendance Screen:** Displays attendance status of recognized students in MS Excel .

Each screenshot clearly represents a specific function of the system and validates the system's successful implementation.

5.4 Back-End Representation (Database Used)

The system uses a **Pickle file (database)** as the backend to store and manage data efficiently. The database ensures data integrity, security, and fast retrieval of student and attendance records. The backend supports real-time updates when a student is recognized and attendance is marked.

5.5 Snapshots of Database with Brief Description

The following data is store inside the local database in the system:

```
[3]: import pickle
with open('data/Entry_numbers.pkl', 'rb') as w:
    Entry_number = pickle.load(w)
    print(Entry_number)
```

[illegible]

```
[1]: import pickle
with open('data/names.pkl', 'rb') as f:
    NAMES = pickle.load(f)
    print(NAMES)
```

[illegible]

```
[5]: with open('data/faces_data.pkl', 'rb') as f:
      FACES=pickle.load(f)
      print(FACES)
```

```
[ [ 47 43 43 ... 19 22 23]
 [131 133 151 ... 62 71 93]
 [ 31 23 30 ... 38 39 49]
 ...
 [ 36 41 46 ... 62 71 98]
 [ 73 74 83 ... 88 99 132]
 [134 140 147 ... 90 102 134]]
```

```
[3]: import pickle
with open('data/Emails.pkl', 'rb') as f:
    EMAILS = pickle.load(f)
    print(EMAILS)
```

[illegible]

✓ 5.5.1 Data store inside the pickle file...

Conclusion

The Face Authentication–Based Attendance System successfully demonstrates an automated, efficient, and secure method for attendance management using facial recognition technology. By integrating Haar Cascade for accurate face detection and K-Nearest Neighbour (KNN) for robust face recognition, the system eliminates common drawbacks of manual and contact-based attendance methods such as proxy attendance, time consumption, and hygiene risks. The implementation confirmed high accuracy rates (above 95%) under varying real-world conditions including lighting and facial expressions. The system's intuitive user interface, streamlined database management, and real-time processing capabilities contribute to operational efficiency in educational purpose. Overall, the project achieved its objectives by delivering a scalable, and user-friendly system .

Future Scope

Future enhancements for this system could focus on:

- Incorporation of deep learning–based face recognition models (e.g., CNNs, FaceNet) to improve accuracy and robustness against pose and occlusions.
- Integration with cloud platforms for centralized database access, enabling remote attendance monitoring and scalability for large institutions.
- Mobile application support for on-the-go attendance marking and notifications.
- Advanced liveness detection techniques to prevent spoofing attacks using photographs or videos.
- Real-time analytics dashboard with AI-powered insights on attendance trends and irregularities.
- Support for multiple concurrent camera feeds to accommodate large exam halls or corporate campuses efficiently.

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